



Mid-Atlantic Fishery Management Council

Comprehensive Five Year (2025–2029) Research Priorities

Approved December 2024

Introduction

The 2006 reauthorization of the Magnuson-Stevens Act (MSA) required that each federal Council develop a five-year research priorities document. The research priorities developed by the Council should address “fisheries, fisheries interactions, habitat and other areas of research that are necessary for management purposes.” NOAA Fisheries and the regional science centers are to consider these research priorities when developing their own research priorities and budgets within the region of the associated Council(s).

The Mid-Atlantic Fishery Management Council (Council), in coordination with the Scientific and Statistical Committee (SSC), completed its first research priorities plan in 2008. That plan was primarily informed by reviewing research recommendations within the various stock assessment documents and the Council’s Research Set-Aside Program. The 2016-2020 research plan was approved in 2015 and the Council’s Visioning Project and Strategic Plan played a critical role in developing and identifying key research themes and elements, many of which are still contained in the updated document. The 2020-2024 research priorities document was approved by the Council in 2019 and developed a year early in order to align with and be informed by the development of the Council’s 2020-2024 Strategic Plan and 5-Year Cooperative Agreement. The entire document was re-organized and re-prioritized to develop a more useful, tactical, and strategic document to effectively advance and apply scientific and management information throughout the Council process. The current document also identified new processes to review, update, and monitor progress to improve the document and help ensure its successful implementation.

In developing the 2025-2029 research priorities document, Council staff followed a similar approach to what was done when developing the current document. Throughout 2024, input was solicited on existing research priorities and potential new priorities from the Advisory Panel, Monitoring Committee and SSC for each species/FMP as part of the fishery specification review process. The SSC also provided feedback regarding the overall document content and updates to existing research priority themes. Council staff, including the staff lead, and the Northeast Fisheries Science Center (NEFSC) assessment lead then reviewed all of the species-specific input received, including new priorities identified during the stock assessment peer review process, and developed the comprehensive priorities list for Council consideration. Staff also reviewed other research, strategic planning, or guidance documents developed by the Council or regional science/management partners and incorporated research relevant priorities into the updated document. These documents included, but were not limited to, the [2025-2029 MAFMC Strategic Plan](#), the [Mid-Atlantic Council's 2024-2027 Inflation Reduction Act projects](#), the [East Coast Climate Change Scenario Planning action menu](#), and the priorities and action items from the [NOAA Climate, Ecosystems, and Fisheries Initiative](#) and recently revised [Ecosystem-Based Fisheries Management Road Map](#).

The 2025-2029 comprehensive research priorities document begins with a review of the current priorities document to evaluate the use and utility of the document to the Council and its regional partners. Updated broad research themes are then included that reflect current information and processes and incorporate SSC input and stakeholder feedback received during the development of the document. Revised and re-prioritized species-specific research lists for Council-managed species are then provided. Lastly, short- and long-term strategies and approaches to improve the document's effectiveness are provided.

Review of Current Five-Year Research Priorities

As mentioned above, the MSA specifies the Council develop a list of research priorities and those lists be provided to NOAA Fisheries and the NEFSC to help inform science and budgeting needs and priorities for the region. Understanding how these research priority documents have been utilized by the Council and the NEFSC in allocating resources to address the identified science and management priorities is particularly

important to understand given potential differences in overall science goals, objectives, and time/funding scales between the Council and NEFSC. During the development of the 2020-2024 document, the SSC questioned how the document is used by the Council and the NEFSC and recommended a review be conducted to determine if the research priorities in the current plan were addressed and if any of the research was used to support stock assessments or within the management process. To evaluate the utility and application of the document by the Council, a review of Council-funded science and management projects was conducted and included for the first time in the current document. This retrospective evaluation of Council-supported activities was found to be a very informative process and one way to quantify the potential benefits and application of the research priority document.

This analysis was updated and a review of Mid-Atlantic Council supported scientific and management projects from 2020–2024 was conducted. Each relevant Council-funded project was reviewed to determine if the project was identified in the current five-year research priorities document – either addressed one, or more, of the seven research priority themes or a species-specific research priority. Each project was also reviewed to determine if the information/outcomes/analysis were used by the Council in the management process or if considered as part of the development of the stock assessment (Table 1).

A total of 23 projects were supported by the Council during this time-period covering eight species and all FMPs. Similar to previous reviews, all of the projects addressed at least one broad priority theme, with many addressing multiple themes. “Stock assessment improvements” (theme A in Table 1) was the most commonly addressed research theme with nearly 57% of all projects expected to help improve and inform Council-managed stock assessments. This was followed by projects that would help address “Climate change impacts on stock productivity and distribution shifts” (theme G in Table 1) which was a new research priority theme added to the 2020-2024 document and identified by the SSC. In addition, 16 of the projects addressed a total of 20 species-specific priorities, or nearly 19% of all species-specific priorities identified in the 2020-2024 research priorities document. Nearly 75% of the Council-funded projects have been, or likely will be in the future, used to support or inform a stock assessment or management action. Some of the projects supported during this time period have either recently finished or are currently on-going, so it is difficult to evaluate their full application to management or stock assessments at this time.

A comprehensive evaluation of the utility and use of the research document by the NEFSC is difficult to conduct and is not included here. However, the [New England and Mid-Atlantic Geographic Strategic Plan 2024-2027](#) developed by the NEFSC and Greater Atlantic Regional Fisheries Office (GARFO) includes a number of priorities and strategies that align with the broad research themes and needs identified in the Council’s five-year priorities document. Common priorities between the Council and NEFSC/GARFO plans include: maintain and improve fishery data collection programs, incorporation of ecosystem level information into stock assessments and management, improving stock assessment information, further developing ecosystem, oceanographic, and habitat modelling approaches, increased utilization and incorporation of social and economic information into the management process, and understanding the economic, biological, and science impacts of offshore wind development. In addition, as part of the Northeast Regional Coordinating Council (NRCC) [stock assessment process](#), research track stock assessments explicitly consider all relevant species-specific research priorities identified by the Council. The stock assessment reports provide information as to how a priority was addressed or rationale as to why it wasn’t. This approach ensures Council research priorities are being considered as part of the stock assessment process to help improve science and management outcomes.

Table 1. Summary of Mid-Atlantic Fishery Management Council supported projects from 2020-2024 used to support science and management needs as identified in the Five-Year (2020-2024) Research Priorities document. Abbreviated research priority themes are: A) Stock assessments, B) Discards, C) Social and economic data, D) Allocation, E) Recreational data, F) Ecosystem tools and EAFM, G) Climate change impacts. @@ indicates project supported a management action or process; ∞∞ indicates project was considered in the stock assessment.

Project Title	Primary Species/FMP	Research Priority Theme	Species Specific Priority
Assessing Current and Changing Marine Fish Habitat and the Spatial Distribution of Key Marine Fish Species along the Northeast Region @@	Omnibus	F, G	#9
Surfclam Species Diagnostics and Population Connectivity Estimates to Inform Management	Surfclam	A, F, G	#1
Summer Founder Recreational Discards Management Strategy Evaluation: Structured Decision Making and Project Facilitator @@	Summer Flounder	B, C, E, F	**
Risk Policy Management Strategy Evaluation @@	Omnibus	C	**
Fishery-Independent Golden Tilefish Bottom Long-Line Survey in Mid-Atlantic (2 surveys during time period) ∞∞	Golden Tilefish	A, B, G	#15, #59, #60
Development of an App for Reporting Recreational Tilefish VTRs	Golden and Blueline Tilefish	A, B, E	#41
Ocean City, Maryland Recreational Effort Video Estimation	Omnibus	A, E	**
Development and Analysis in Support of the <i>Illex</i> Research Track Stock Assessment ∞∞	<i>Illex</i> Squid	A	#8, #67
Evaluating the Importance of Atlantic Chub Mackerel in the Diet of Highly Migratory Species in the Northwest Atlantic	Chub Mackerel	A, F	#11
Piloting a Low-Bycatch Commercial Squid Jig Fishery in Southern New England	Longfin Squid	B	#72
Golden Tilefish IFQ Program Review @@	Golden Tilefish	C, D	**
Testing and Application of a Length-Based Assessment Model for U.S. East Coast Shortfin Squid ∞∞	<i>Illex</i> Squid	A, G	#8, #67, #68, #69
Technical Support and Analysis for the Longfin Squid Research Track Stock Assessment ∞∞	Longfin Squid	A, G	#8, #73, #77, #78
Spiny Dogfish Aging for Research Track Stock Assessment ∞∞	Spiny Dogfish	A	#10, #96
Preparation, processing, and aging of the appropriate hard part samples for MAFMC managed species ∞∞	SF, Scup, BSB, A. Mackerel, Butterfish, Golden Tilefish, Bluefish, SCOQ	A	#10
Advancing Inshore Fish Habitat Modeling for the Northeast Regional Habitat Assessment @@	Omnibus	F, G	#9
Northern Expansion of South Atlantic Deepwater Longline Survey (2 surveys during time period) ∞∞	Golden and Blueline Tilefish	A, G	#9, #15, #42, #45
Longfin Squid Biological Sampling ∞∞	Longfin Squid	A	#8, #74

Longfin Squid Ageing ∞∞	Longfin Squid	A	#74, #78
Mid-Atlantic Council Program Review	Omnibus	G	**
Northeast Biological Port Sampling Program ∞∞	Omnibus	A	**
Evaluation of Low-Profile Gillnets to Mitigate Sturgeon Bycatch in the Commercial Spiny Dogfish Fishery	Spiny Dogfish	B	**
Tilefish Reporting and Permitting Outreach and Evaluation	Golden and Blueline Tilefish	E	#41

Research Priority Themes

Key research themes are included to address broad concepts that cut across a number of Council-managed species. These themes are responsive to the Council's updated (2025–2029) Strategic Plan¹ regarding the science and ecosystem information used in the management process. The Science goal in the 2025-2029 Strategic Plan focuses on the core of the Council's mandated science-based decision-making process and seeks to improve data accuracy, credibility, and the use of collaborative research in the science and management process. The Ecosystem goal that specifies the Council *support the ecologically sustainable utilization of living marine resources in a manner that maintains ecosystem productivity, structure, and function*. This goal seeks to address a wide range of Council issues related to climate change, forage, habitat, species interactions, and other factors that impact the health of the marine ecosystem. Aligning the Council's research priorities with the Strategic Plan ensures consistency, appropriately prioritizes Council resources, and improves coordination of science and management efforts throughout the region.

A. Stock assessment improvement

Improvements to the data and analysis supporting the stock assessment process continue to be identified as the Council's top research priority and the SSC strongly recommended the continued focus on stock assessment improvements in this edition as well. The NEFSC has invested significant resources in transitioning assessments to a more robust and flexible state-space modeling framework², the Woods Hole Assessment Model (WHAM). The transition to WHAM has resulted in significant stock assessment improvements for a number of Council managed species including black sea bass, butterfish, bluefish, and golden tilefish and there are near-term plans to implement WHAM for Atlantic mackerel and scup. In addition, the NRCC stock assessment process created two types of stock assessments (management track and research track) that provided for greater assessment flexibility and quality, increased research opportunities and established a long-term assessment schedule. The process has allowed for applied stock assessment research, more timely stock assessment information, and helped contribute to the significant advancements in the regions stock assessment capabilities and capacity.

While science advancements have been made and new information obtained to support stock assessment improvements (see Table 1 for examples), the region has experienced some erosion of the fishery independent and dependent data collection efforts that are foundational for robust stock assessments. Therefore, a continued focus and advancement of data collection programs that improve size/age composition of the catch, discard estimates and associated mortality rates, and fishery independent abundance information remains a priority. Feedback obtained during the development of the 2025-2029

¹ For more information about the Mid-Atlantic Council's 2025-2029 Strategic Plan, please see - <https://www.mafmc.org/strategic-plan>.

² For more information about state-space models and the research track stock assessment peer review on their application to Northeast stocks, please see - <https://www.fisheries.noaa.gov/event/applying-state-space-models>.

Strategic Plan also highlights the need for continued science-based industry collaboration and increased utilization of fishing fleet information and on-water observations. In addition, recent research track stock assessments have included the development of the Ecosystem and Socioeconomic Profiles (ESPs) that provide a broad range of ecosystem and socioeconomic information at the species-specific level. The continued development and inclusion of ecosystem information (e.g., ESPs) into the management process and environmental covariates into stock assessments remain a priority.

B. Research to support measures which reduce/eliminate discards

Obtaining accurate discard information and the management challenges to reduce regulatory discards remain, particularly within the recreational sector. Stakeholder feedback during the development of the strategic plan and during many Advisor Panel meetings focus on the need significantly reduce discards and develop new management strategies to convert regulatory discards into harvest to provide both economic and biological benefits. Reducing regulatory discards through improved gear performance, and the development of management procedures and approaches to allow for greater retention of catch or the avoidance of unmarketable, sub-legal or otherwise prohibited species should continue to be explored.

The Council has supported a variety of discard related projects ranging from potential gear and time/area closure modifications in the commercial summer flounder, scup, black sea bass, and spiny dogfish fisheries to [identifying strategies to reduce discards](#) and increase opportunities in the recreational summer flounder fishery. However, findings from those projects have yet to directly change management approaches and additional research, data collection and management strategies are needed. There is also a need for continued focus on collaborative research opportunities with both commercial and recreational vessels to evaluate gear selectivity, discard mortality estimates, and innovative management strategies to avoid and minimize discards.

C. Collect and incorporate social and economic data into fishery management decision process and stabilize yields

The continued collection, analysis, and increased utilization of social and economic information in the Council's decision process remains a high priority for the Council, the SSC, and stakeholders. While the Council has been successful in meeting the biological mandates of the MSA, the resulting social and economic consequences have been viewed as unnecessarily severe by both commercial and recreational stakeholders. Unfortunately, basic information on the number of fishermen and their permits, the associated costs to determine profitability of vessels in a port, and how profits change with regulatory changes, is often limited.

Over the last several years, the Council and NEFSC have developed a range of tools and indicators to help integrate socioeconomic information into policy and management actions. For example, the Council has made significant Ecosystem Approach to Fisheries Management (EAFM) advancements including a comprehensive review and update to the EAFM risk assessment which includes nine different social and economic risk elements that may threaten achieving the social and economic objectives the Council may have for its fisheries. Similarly, the Mid-Atlantic State of the Ecosystem report developed by the NEFSC begins with a range of ecosystem indicators that describe the potential drivers and implications to meeting the Council's socioeconomic management objectives for seafood production, commercial profits, recreational fishing opportunities, and community vulnerability. The development of Ecosystem and Socioeconomic Profiles through the research track assessment process and the NEFSC Fishing Community Climate Vulnerability Assessments also provide for stock and community-specific social and economic information that can support management decisions. However, further socioeconomic indicator development and simulation testing through management strategy evaluation is needed to consider the socioeconomic

implications and trade-offs associated with different management and policy choices for Council-managed species.

The majority of the social and economic information available for the commercial sector is collected through voluntary surveys with permitted vessels, dealers, and processors. Participation in these voluntary surveys has declined for many Mid-Atlantic fisheries, resulting in less socioeconomic information available to understand and evaluate changes in fleet dynamics and profitability. New or additional data collection programs need to be developed in collaboration with the fishing industry to help ensure buy-in and trust in providing this type of information. Highlighting the need, utility, and benefits of providing this information can help alleviate some industry concerns and promote support for these types of data collection efforts.

Similar challenges are associated with the collection and application of economic information for the recreational sector. The information is generally considered uncertain and when economic information is collected, it's typically done as one-off analysis and infrequently updated, limiting the ability to inform management in a timely manner. In 2023, NOAA Fisheries held a recreational constituent [workshop](#) which developed a suite of recommendations to support the collection, analysis, and uptake of recreational economic information into the science and management process. Recommendations included the increased use of electronic technologies such as smartphone apps, greater collaboration and partnerships with management, scientists, and the recreational industry to identify opportunities to collect and share timely and proactive economic data, and dedicating resources to estimate optimum yield for recreational fisheries given varying angler preferences (harvest versus catch and release) for certain species. The Council should continue to work with NOAA Fisheries and engage with the recreational community to identify opportunities to implement these recommendations.

D. Evaluation of existing allocations to fishery sectors

A number of Council managed species allocate the Acceptable Biological Catch (ABC) by sector (commercial/recreational) and, in some cases, by state or permit type. The fairness, equity, and overall management structure of many of the current allocation scenarios have been questioned by stakeholders and fishery managers. In addition, stakeholders have noted the general inflexibility of the fixed quota allocation system currently in place and recommended that the Council consider alternative methods to allocate annual catch limits. Changing species distributions, stock productivity and continual updates to the Marine Recreational Information Program (MRIP) catch timeseries have only added to the need to evaluate and consider allocation scenarios that are robust to these changing conditions.

The Council has been working to address these different allocation challenges and recent Council actions (e.g., Summer Flounder Commercial Issues Amendment and the Black Sea Bass Commercial Allocation Amendment) have considered or included dynamic allocation strategies to incorporate changing stock distributions in specifying revised allocations. The Council is also considering potential allocation strategies within the recreational sector (i.e., private versus for-hire) to account for changing angler behavior and fishing opportunities. However, not all stakeholders have been supportive of the efforts and outcomes to date and more allocation decisions will need to be addressed in the future. Therefore, the need to identify methods and analyses (e.g., management strategy evaluation and scenario planning) that can test, evaluate, and identify alternative management strategies and determine optimal allocation options that incorporate biological, social and economic considerations remains a high priority.

E. Recreational data collection and utilization

The SSC continues to highlight the need for and importance of recreational data collection as a priority research theme. Future improvements and revisions to the MRIP recreational catch timeseries are anticipated. These changes will need to be incorporated and evaluated in upcoming stock assessments and

there will be continual implications within the management system that will need to be addressed by the Council. The SSC has noted the inherent uncertainty and variability of recreational data and future large-scale changes to the MRIP recreational catch timeseries represent an important source of scientific uncertainty when developing ABC recommendations.

The Council has worked to reduce the uncertainty of recreational information and improve recreational data collection efforts for Council-managed fisheries. For example, the Council collaborated with the NEFSC, members of the SSC, and stakeholders to develop recreational demand models that can account for changes in stock abundance and angler behavior and are now used to help set recreational management measures. As part of the [Recreational Measures Setting Process Framework/Addenda](#) development, a management strategy evaluation was conducted by the NEFSC and members of the SSC to evaluate the biological and socioeconomic impacts and trade-offs associated with different recreational management alternatives. Finally, the Council implemented mandatory permitting and reporting for the recreational blueline and golden tilefish fisheries in order to collect comprehensive recreational landings and effort information. While additional work, analysis, and evaluation is needed for all of these activities, they represent significant advancements in recreational data collection and utilization efforts.

Modernizing Recreational Fisheries Management Act of 2018 requires the increased incorporation of various recreational data sources and an evaluation of alternative data collection methods (e.g., smart phone apps and other electronic reporting options). While some advances have been made in the use of data collected from smart phone apps, particularly as it relates to the impacts of the offshore wind development, additional tools and analyses are needed to begin to integrate this information more broadly into science and management processes. Developing a recreational study fleet program in the Northeast to collect fine scale recreational data was a high priority recommendation from the East Coast Climate Change Scenario Planning project.

This theme also looks to not only advance new and additional recreational data collection programs to support Council activities, but to also develop new and alternative methods to evaluate and incorporate recreational data into the management process. The use of management strategy evaluation and the continued development of recreational bio-economic modeling approaches have the potential to significantly improve recreational management approaches and should continue to be perused.

F. Collect ecosystem data and development of ecosystem tools and management strategies to support EAFM and “climate ready” initiatives

The Council’s the [Ecosystem Approach to Fisheries Management \(EAFM\)](#) goal is to “manage for ecologically sustainable utilization of living marine resources while maintaining ecosystem productivity, structure, and function”. Climate change is having a profound impact on our fisheries, communities, and management capabilities. To achieve the EAFM goal, the Council will need to better understand and adapt to the challenges of climate change by building a “climate ready” management process to ensure resilient and sustainable fisheries. The Council’s 2025–2029 Strategic Plan and the NEFSC/GARFO New England and Mid-Atlantic Geographic Strategic Plan 2024-2027 identify ecosystem science as a major goal, theme or strategy. There is broad support for the continued collection of ecosystem-level climate, habitat, fleet dynamics, and species interaction information to help improve our understanding on the current and anticipated impacts of climate change on the region’s fisheries and the broader marine ecosystem.

Since the approval of the EAFM Guidance document in 2016, the Council has worked to implement and advance a variety of management actions and science activities associated with the four ecosystem priorities of forage, habitat, climate change/variability, and ecosystem level interactions. The NEFSC, SSC, and Council continue to refine ecosystem indicators, develop ecosystem models, and conduct simulation testing that

have not only benefited the Council's EAFM initiatives, but have been integrated into the Council's more traditional management products and processes. Advances in scientific information will lead to the continued improvement, development, and utilization of ecosystem tools, products, and processes such as Integrated Ecosystem Assessments, State of the Ecosystem reports, Ecosystem and Socioeconomic Profiles, Climate Vulnerability Assessments, and Climate-Ready Fisheries Management. The future success of the Council's EAFM process relies on the continued support and investment in ecosystem science and data collection, analytical tools, and management strategies. In addition, the Inflation Reduction Act (IRA) funding for NOAA and the subsequent IRA funding provided by NOAA Fisheries to the Council provides an unprecedented opportunity to build on the Council's efforts and make significant advancements in support of climate ready science and management.

G. Climate change impacts on stock productivity and distribution shifts

Climate-related changes in the Mid-Atlantic have already been widely observed and documented by fishermen, managers, and scientists. These changes in the environment have led to shifts in stock distributions, possible changes in stock productivity and have the potential to impact the Council's ability to effectively manage these resources. Climate induced changes to ocean acidification, food web dynamics, and habitat can also affect growth, natural mortality, and fecundity which can also have implications for stock productivity. While this research theme is embedded in a number of the other included themes (e.g., stock assessment, socioeconomic considerations, allocation, and EAFM initiatives), the SSC recommended it be a stand-alone theme given the importance of this issue and its linkages to other research and management priorities. Incremental scientific advances under this theme can inform efforts and activities under other priority themes. Enhanced data collection programs to detect change and the development of short/mid-range distribution forecast models, such as [MOM6](#) and dynamic range models, to understand the drivers and magnitude of change and the associated biological and management risks are critical research needs. Developing management strategies and governance structure options through MSE simulation, scenario planning and/or structured decision making are necessary to create adaptive approaches to respond to continually changing conditions and risks. By taking advantage of the East Coast Climate Change Scenario Planning project, the Council's EAFM structured decision framework and the IRA funding/projects, the Council is well positioned to develop the tools necessary to understand and respond to changing stock conditions and distribution shifts.

Species Specific Priorities List

Similar to the approach taken with the current document, a broader and more comprehensive process to solicit input on research priorities was undertaken for this document. Input on current and new priorities was provided by the Advisory Panel, Monitoring Committee, and the SSC as part of the specification review/setting process for each Council-managed species. Staff then worked with the Council species lead and the NEFSC assessment lead to review all input received, as well as the research priorities identified in the research and management track stock assessment reports and SSC meeting reports, to develop a revised list of species-specific research priorities. It is important to note that these lists are not meant to be exhaustive and cover every issue, science need or management topic that has been raised for a particular species. These lists are meant to focus on some of the more critical and important areas of consideration to advance science, stock assessment approaches and results, and improve management outcomes.

Research priorities are separated into two different categories, short-term/smaller scale and long-term/larger scale projects. Within each category, the different research topics are then listed in priority order. This type of approach was suggested by the SSC and as a way to reflect the different end users of this document – the Council, the NEFSC and other science partners – and to devise a document that is both

tactical and strategic in addressing the most important research and science needs for effective management by the Council. The short-term/smaller scale priorities provide a tactical approach to answer specific scientific and management questions, particularly when limited resources (i.e., funding, expertise, and staff) are available. It should be noted that the use of the term “scale” to describe and categorize priorities does not refer to spatial or geographic scale, but references the size and scope of a particular priority. A short-term/smaller scale priority could be large in spatial/geographic scale but focus on a specific question in which data collection and research could be completed in a short period of time with less resources needed to complete. These priorities are where the Council would likely focus its attention and are the types of projects the Council has typically supported in the past when opportunities are available. Addressing these short-term/small scale projects can lead to incremental advances in support of long-term/larger scale priorities that are more strategic and seek to address larger concepts and issues that likely require significant resources over an extended period of time. This approach allows the Council, NEFSC and other partners to leverage resources, for example matching funds and technical expertise, to identify funding opportunities to address these larger projects.

Below is the updated comprehensive list of research priorities for each Council-managed species, in alphabetical order. In addition to the species-specific lists, there is also a list of research priorities that are more general and/or have applicability across several or all Council-managed species. For example, priorities related to habitat, socioeconomic information, allocation strategies and stock structure dynamics are topics that are covered in this section. Lastly, in order to ensure an individual research priority addresses seven broader priority themes identified by the Council, the corresponding theme(s) associated with each research priority are identified in column on the right.

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Comprehensive list of research needs for Mid-Atlantic Council managed species (## - work has been done, is currently ongoing, or in development to address priority)

GENERAL OR CROSS-SPECIES	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
1. Investigate stock structure utilizing otolith microchemistry, genetic analyses, or other approaches for different Mid-Atlantic stocks (e.g., golden and blueline tilefish, black sea bass, Atlantic and chub mackerel).	A, F, G
2. Understand the fishery objectives and performance measures for the fishery from a biological and socioeconomic perspective, to evaluate the balance of costs and benefits of ABC specifications (e.g., variable vs. average ABC).	B, C
3. Explore and develop tools and opportunities to collect and utilize citizen science and local ecological knowledge into stock assessments and other science and management products.	C, F, G
4. Create a framework to improve social science information regarding crew and shore-side employment, remuneration and job satisfaction for all Mid-Atlantic fisheries.	C
5. Evaluate expanded use of data collected by the industry study fleet for all Mid-Atlantic stocks. ##	A, B, F, G
6. Evaluate the relationship between changes in landings limits and the rates and magnitude of discarding in the commercial and recreational fisheries.	B, C, D, E
7. Evaluate the potential to develop and integrate fishery-dependent CPUE indices of abundance into stock assessments. ##	A
8. Explore future use of data loggers as a tool to enforce soak time restrictions in gillnet fisheries.	B
LONG-TERM/LARGER SCALE	
9. Monitor changes in distribution and habitat use, including the timing of spawning and migration or other ecological changes, and the implications to stock productivity and governance for all Mid-Atlantic species. ##	A, B, D, F, G
10. Collect accurate size and age composition of commercial and recreational catch (including the discarded component of the catch) to develop or improve catch at age matrices for all managed stocks.	A, B, E
11. Collect, synthesize, and incorporate climate, oceanographic, and ecosystem level information into single and multi-species assessment and management models. ##	A, F, G
12. Investigate potential sector and regional allocation changes and adaptive management strategies to respond to changing environmental conditions.	C, D, F, G
13. Evaluate and monitor the impacts of all phases of offshore wind energy activities, including cumulative impacts and impacts on fisheries-independent data collection. ##	A, F, G
14. Develop tools to collect representative economic information on fixed and variable trip costs to understand fleet profitability for all Mid-Atlantic fisheries.	C, E, F
15. Evaluate and monitor socioeconomic impacts of offshore wind energy development on Council-managed fisheries, including changes in fishing behavior, changes in the distribution of fishing effort, changes in revenues, differential impacts on commercial and recreational fisheries, impacts to shoreside businesses and communities, and cumulative impacts. ##	C, E, F
16. Develop a recreational study fleet program in the mid-Atlantic to support the assessment and management process.	A, B, G, E

17. Implement or continue efforts to implement novel or alternative surveys to derive fishery independent indices of abundance (black sea bass, blueline and golden tilefish, Atlantic and chub mackerel). ##	A
18. Explore effectiveness of low-profile gillnet gear in gillnet fisheries as a potential future management tool to reduce the bycatch of Atlantic sturgeon. ##	B, F, G

ATLANTIC MACKEREL	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
19. Continue to collect and evaluate mackerel egg data (ECOMON survey). ##	A
20. Investigate stock structure, mixing, and spawning differences of the two stock contingents based on recent otolith microchemistry and/or genetic data.	A, F, G
LONG-TERM/LARGER SCALE	
21. Evaluation of time and age-variant natural mortality and predation mortality for this stock.	A, F, G
22. Develop methods for using acoustics to determine Atlantic mackerel abundance and/or catchability.	A
23. Initiate a reproductive study in the U.S. to obtain fecundity estimates and spawning seasonality. Update Canadian fecundity estimates (which are currently based on a 1986 publication) and compare estimates between countries.	A
24. Obtain biological samples from all components of the fishery and covering both spawning contingents.	A

BLACK SEA BASS	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
25. Evaluate the biological, management, and socioeconomic drivers of recreational and commercial harvest and discards.	B, C, E
26. Improve the understanding of discard mortality from various gear types, considering differences in location and time of year, particularly for gear types with limited recent research (e.g., gillnets and trawls).	A, B
LONG-TERM/LARGER SCALE	
27. Collect fishery dependent and independent data at appropriate spatial and temporal resolution to inform patterns in growth, recruitment, and mortality.	A, G
28. Enhance fishery dependent data collection across all size and market categories to improve data that support estimation of fishery length and age compositions.	A
29. Investigate environmental drivers of recruitment, including the application to the stock assessment and projections. ##	A
30. Evaluate the stock structure and spatial overlap of black sea bass between the mid and south Atlantic. ##	A, F, G
31. Investigate the impact of a changing environment due to climate change on the life history and spatial dynamics of the stock and fisheries.	A, F, G

BLUEFISH	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
32. Develop a Recreational Fleet Dynamics Model (RFDM) for bluefish to estimate recreational catch and evaluate the biological and management drivers of recreational harvest and discards.	B, E
33. Evaluate changes in selectivity of age-0 bluefish in fishery independent surveys due to shifting environmental conditions. Investigate trends in recruitment. ##	A, G
34. Conduct a post-release mortality study to determine if the recreational discard mortality rate has changed over time.	A, B, E
LONG-TERM/LARGER SCALE	
35. Develop a fishery independent index and/or fishery dependent sampling program of offshore populations of bluefish to capture larger, older fish.	A, G
36. Investigate how environmental variability may affect timing of migration patterns of juvenile bluefish and the distribution of adults, which in turn, may affect availability. ##	A, G
37. Develop a coordinated tagging program to help understand migration patterns potentially contributing to patterns in length frequency distributions.	A, G

BLUELINE TILEFISH	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
38. Identify data sources and sampling methods to improve the biological length samples of commercial and recreational landings to better characterize the size distribution of removals.	A, E
39. Consider enhancements to improve mandatory electronic reporting for all recreational anglers and collect fishery-dependent information such as effort, total catch and length information on harvested and discarded fish. ##	A, B, E
LONG-TERM/LARGER SCALE	
40. Investigate new stock assessment approaches, including dynamic non-equilibrium methods.	A
41. Research the reliability of aging methods and determination of growth parameters (e.g., intensive tagging survey).	A
42. Collect additional biological samples to enhance understanding of life history dynamics and biological characteristics of the stock (e.g., age and size of maturity, maximum age, fecundity, spawning periods). ##	A
43. Conduct habitat studies of deep-water sites in the mid-Atlantic (Norfolk Canyon, Baltimore Canyon, and Hudson Canyon).	A, G

BUTTERFISH	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
44. Examine the efficiency (including day vs. night) of survey gear and potential changes in butterfish catchability including a parallel catchability estimate for NEFSC Spring surveys so that both Spring and Fall surveys can be included in the model.	A
45. Examine reference points based on total biomass and forage-specific reference points using natural mortality.	A
45. Further investigate the methods used to identify changes in recruitment stanzas.	A
47. Evaluate the uncertainty in the ad hoc F_{MSY} proxy and effects on catch advice.	A
LONG-TERM/LARGER SCALE	
48. Further investigate methods to inform population scaling within assessments.	A

49. Further investigate the role of butterfish in the ecosystem and refine predation mortality estimates. ##	A, F
50. Reconsider stock structure and degree of exchange with south Atlantic stock component (i.e., stock ID).	A, G

CHUB MACKEREL	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
51. Develop expanded discard estimates for the region and better quantify South Atlantic catch.	A
52. Collect length, age, growth, maturity information from fishery independent and dependent data sources throughout U.S. Atlantic waters.	A
53. Evaluate catch per unit effort including the influence of environmental and socioeconomic factors.	A, C, G
54. Investigate existing egg and larval surveys throughout the U.S. Atlantic coast to better understand chub mackerel recruitment dynamics.	A
LONG-TERM/LARGER SCALE	
55. Investigate stock mixing throughout Atlantic waters, as applicable.	A
56. Investigate habitat use at different life stages.	A, F

GOLDEN TILEFISH	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
57. Continue to utilize fishery-independent information to assess whether the dome-shaped selectivity curve used in the assessment reflects fishery selectivity or availability, or both. ##	A
58. Develop sampling programs to increase information of recreational landings at size and age.	A, E
59. Continue to assess the accuracy and reliability of aging techniques.	A
LONG-TERM/LARGER SCALE	
60. Explore the development of alternative data streams to better capture the size distribution of the fish caught in this fishery, such as electronic monitoring onboard vessels (to collect length data) or electronic size monitoring programs (to collect paired length and weight) at processing facilities.	A
61. Identify golden tilefish spawning locations through cooperative research aboard commercial fishing vessels to collect coincident oceanographic and biological (egg and larvae) data.	A, G
62. Evaluate data collection methods to increase information on gear/ species interactions (i.e., pot/trap, spiny dogfish) and bait type to understand their effects on the commercial CPUE index.	A, B, F
63. Evaluate the Woods Hole Assessment Model (WHAM) performance for information poor stocks using simulated tilefish like populations (i.e., only catch data).	A
64. Continue to evaluate the effects of climate and environmental indices on stock dynamics for the potential development of environmental covariates to be included in the stock assessment.	A, F, G
65. Evaluate the role of the golden tilefish gear restricted areas on the stock and its fisheries.	A, F

ILLEX SQUID	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
66. Continue to increase availability of fine scale fishery data to track within season changes in the composition and length structure of <i>Illex</i> . ##	A
67. Refine our understanding of stock structure and distribution of <i>Illex</i> to U.S. surveys and fishery, resulting from long-term changes in climate or other oceanographic factors.	A, F, G
LONG-TERM/LARGER SCALE	
68. Improve our understanding of <i>Illex</i> biology and life history (e.g. reproduction, growth, and mortality) in the context of changing environmental conditions.	A, F

LONGFIN SQUID	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
69. Further develop practicable ways to reduce bycatch.	B
70. Refine understanding of availability and catchability in surveys (e.g., fall NEAMAP-Bigelow comparisons and conversion factors).	A
LONG-TERM/LARGER SCALE	
71. Until real-time assessment is feasible, expand cohort analysis to refine stock assessments. and their incorporation of seasonal indices (currently spring and fall are just averaged). ##	A
72. Continue to increase the availability of age and growth studies to better estimate growth, reproduction, and mortality patterns and to discern seasonal productivity/catchability patterns. ##	A
73. Evaluate methods of incorporating ecological relationships, predation, and oceanic events that influence abundance and availability. ##	A, F
74. Refine understanding of stock range, distribution, and structure and availability to U.S. surveys and fishery.	A, G

OCEAN QUAHOG	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
75. Conduct research to better understand life history for an extremely long-lived species at appropriate temporal and spatial scales (growth, size-at-age, recruitment, natural mortality, maturity-at-length, and fecundity – in order of priority).	A
76. Evaluate the cost and benefit of different technological methods (e.g., HABCAM, EM, AI, and optical surveys) for measuring ocean quahog abundance, length frequency, and habitat. ##	A, F
LONG-TERM/LARGER SCALE	
77. Conduct work to support spatially explicit stock assessments that account for source and sink differences in productivity (i.e., are some areas more important to productivity than others).	A
78. Development of techniques to age ocean quahogs in a cost-effective manner. ##	A

SCUP	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
79. Investigate the potential causes and implications of the declining mean-weight-at-age in recent years.	A, G
80. Evaluate the spatial and temporal overlap of scup and squid to better understand and characterize fishing effort and identify scup discard/bycatch patterns and hotspots. ##	A, B, F
81. Explore the regulatory and economic factors driving scup discards and evaluate the biological and economic trade-offs of management strategies to reduce scup bycatch. ##	B, C
82. Characterize the pattern of selectivity for older ages of scup in both surveys and fisheries.	A
LONG-TERM/LARGER SCALE	
83. Continue to evaluate the role and relative importance of implemented management strategies and consider the long-term climate variability to the increases in stock abundance and high recruitment events since 2000. ##	A, B, D, F, G
84. Explore climate and ecosystem factors impacting scup availability and productivity and integrate environmental data into the stock assessment model.	A, F, G
85. Characterize the current scup market and explore the development of new markets.	C

SPINY DOGFISH	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
86. Continue to explore model-based methods, such as VAST or other spatial approaches, to derive survey indices for spiny dogfish.	A
87. Explore the sensitivity of the Stock Synthesis 3 model parameterization, configuration, and starting point assumptions.	A
LONG-TERM/LARGER SCALE	
88. Investigate opportunities to increase domestic and/or international market demand.	C
89. Explore the spatial distribution of spiny dogfish beyond the current NEFSC trawl survey geographic footprint (e.g. off-shelf abundance).	A, G
90. Continue to collect, process, and age-spiny dogfish spines to monitor changes in growth over time and support future age-based assessments. ##	A, G
91. Investigate the role of ecosystem effects on spiny dogfish maturity, natural mortality and other life history parameters.	A, F, G

SUMMER FLOUNDER	Corresponding Theme(s)
SHORT-TERM/SMALLER SCALE	
92. Collect length, weight, and age data by sex to fully evaluate the sex and size distributions of landed and discarded fish in the summer flounder fisheries.	A, B, E
93. Evaluate summer flounder recreational discard survival under different environmental variables and gear configurations with survey design considerations that account for feeding and predation.	A, B, E
LONG-TERM/LARGER SCALE	
94. Evaluate range expansion and/or changes in distribution and migration patterns and their implications for stock assessment and management.	A, F, G
95. Continue to evaluate the causes for decreased recruitment, changes in recruitment distribution, and changes in the recruit-per-spawner relationship in recent years. Develop	A, F, G

studies, sampling programs, or analyses to better understand how and why these changes are occurring, and the implications to stock productivity.	
96. Explore the potential mechanisms for recent slower growth (decreasing mean length, weight at age, and maturity at age) that is observed in both sexes.	A, F, G
97. Incorporate sex-specific differences in size-at-age into the stock assessment.	A
98. Evaluate the causes and consequences of summer flounder declines in Chesapeake Bay.	A, F, G
99. Reconsider stock structure based on modern approaches.	A, F, G

SURFLAM	Corresponding Theme(s)
SHORT-TERM/SHORTER SCALE	
100. Conduct research to better understand life history at appropriate temporal and spatial scales (growth, size-at-age, recruitment, natural mortality, maturity-at-length, and fecundity – in order of priority).	A
101. Evaluate the cost and benefits of different technological methods (e.g., HABCAM, EM, AI, or optical surveys) for measuring surfclam abundance and habitat, including clam density. ##	A, F
LONG-TERM/LARGER SCALE	
102. Examine the effects of climate change on the spatial distribution of clams, on the operation of the fishery, and patterns of discarding/incidental mortality, and on the overall productivity of the stock.	A, B, F, G
103. Evaluate small-scale surfclam patch density and the connectivity of the two stock areas (Mid-Atlantic and Georges Bank) and the implications on stock dynamics, particularly reproductive success and recruitment exchange.	A

Future Direction

The current document outlined a new approach and process to evaluate the utility and implementation of the research priorities document. Instead of a biennial review (i.e., two reviews that occur in years two and four, during the five-year period) as used in prior years, a mid-cycle review (i.e., one review during year three of the five-year period) of the current list of species-specific priorities will occur. This type of [review](#) was found to be highly effective and supported by the Council when conducted on the current research priorities document in 2021. This review includes input from the Advisory Panel, Monitoring Committee and SSC to ensure the document is reflective of the current state of scientific knowledge and the Council's science and management priorities. Input on current or new priorities will occur as part of the Advisory Panel development of the Fishery Performance Report and when the SSC and Monitoring Committees review or develop new catch specifications. The review will also track the progress toward addressing research priorities and to identify what research has been completed and why other areas may not have been addressed. It should be noted that this review will not apply to the broader research priority themes which would remain the same for the entire five-year document period.

Lastly, a more comprehensive review and evaluation of the various research plans and priorities across the region (Mid-Atlantic Council, New England Council, Atlantic States Marine Fisheries Commission) should be conducted in the future. All regional management partners develop their own research priority documents with hundreds of research projects identified which compete for limited fiscal and NEFSC resources. A more comprehensive and holistic review can help identify research similarities, highlight differences, and ensure continued communication and coordination to maximize and leverage limited staff and fiscal resources. Given the recent focus and development of climate ready fisheries and management strategies, there is an opportunity to make meaningful progress in regional coordination by utilizing the numerous cross-Council IRA projects, the development and application of NOAA's Climate, Ecosystems and Fisheries Initiative products, and the recent formation of the [East Coast Climate Coordination Group](#). This more holistic review and greater consideration of regional priorities could lead to the development of a comprehensive research priorities [plan](#) for the Council and provide an approach to effectively and efficiently carry out and address the identified research needs identified in this document.

These enhancements, planned reviews, and comprehensive research plan development are included as strategies in the Council's updated 2025–2029 Strategic Plan. Aligning the Strategic Plan and Five-Year Research Priorities will help ensure the Council achieves its science goal and associated objectives.